

Enhancing knowledge tracing through course map and question difficulty analysis

Jianing Xia^a, Amanda Li^b, Hui Yin^c, Guozhe Jin^{d,*}

^a Universal Higher Education, Melbourne, Australia

^b Victoria University, Melbourne, Australia

^c Swinburne University of Technology, Australia

^d Institute of Intelligent Information Processing, Yanbian University, China

ARTICLE INFO

Keywords:

Intelligent education

Knowledge tracing

Concept map

Difficulty effect

ABSTRACT

Knowledge tracing (KT) has enhanced students' learning efficiency in online systems by modeling how students learn over time based on their interactions with educational content. It has gained significant research attention recently due to its crucial importance in intelligent education. Nonetheless, current knowledge tracing (KT) methods have not adequately explored and utilized the impact of concept relations and question difficulty level on students' knowledge state during learning. This study aims to address this gap by investigating the effect of the concept map and question difficulty on learning and proposing the Course Map and Difficulty-enhanced Knowledge Tracing (CMDKT) model to enhance students' knowledge state assessment. Experiments on ASSIST09 and DBE-KT22 show CMDKT improves predictive performance over strong baselines (e.g., +2.47% AUC and +2.32% ACC on average across datasets), while also reducing MAE and RMSE compared to DKT and GKT.

1. Introduction

Due to advancements in machine learning and the increasing availability of educational data, knowledge tracing, as the core technology of intelligent educational services, has gained significant attention and adoption in recent years. Knowledge tracing analyzes learners' past learning performance on various questions, tracks their knowledge states, and quantitatively diagnoses and predicts their current mastery of different concepts. Through knowledge status measurement, students can get timely learning feedback. This allows them to identify any knowledge gaps and the intelligent education system can formulate personalized education plans. Therefore, knowledge tracing models can enhance students' educational experience by predicting their future learning outcomes and offering personalized interventions. These models are also essential to customized learning and adaptive learning systems because they let teachers tailor their instruction according to the needs of each student.

The left section of Fig. 1 illustrates a simplified instance of Knowledge Tracing (KT) employed in student learning scenarios. Here, students participate in practice sessions where they tackle various questions to enhance their understanding. KT models are designed to monitor and assess the progression of students' knowledge states by analyzing the questions they encounter and the corresponding answers

provided during each practice iteration. In the literature, various KT models are employed to track the progression of students' knowledge. Bayesian Knowledge Tracing (BKT) [1] relies on probabilistic inference to estimate students' knowledge state. Performance Factor Analysis (PFA) [2] employs statistical techniques to analyze students' performance on relevant factors to determine their knowledge state. Deep Knowledge Tracing (DKT) [3] utilizes Recurrent Neural Networks (RNNs) to capture the intricate cognitive processes involved in student learning.

However, traditional knowledge tracking methods often oversimplify question records by representing them as sequences of knowledge concepts (KC), without considering the relationships between questions or KCs. As in the right part of Fig. 1, the KC in question Q1 related to "Function", serves as a prerequisite concept for question Q4. Given that the outcome of Q1 can provide valuable insights for predicting the accuracy of responding to Q4. Moreover, existing KC methods do not distinguish between the knowledge acquisition of two students who solve different questions related to the same KC but with different difficulty levels. For example, the student who solves the easier question, Q2, should have a lower knowledge state regarding the corresponding KC (i.e., Inequality) compared to the student who solves the more difficult question, Q3. To overcome these challenges,

* Corresponding author.

E-mail addresses: Jennie.X@uhe.edu.au (J. Xia), amanda.li@vu.edu.au (A. Li), huiyin@swin.edu.au (H. Yin), jinguoze@ybu.edu.cn (G. Jin).

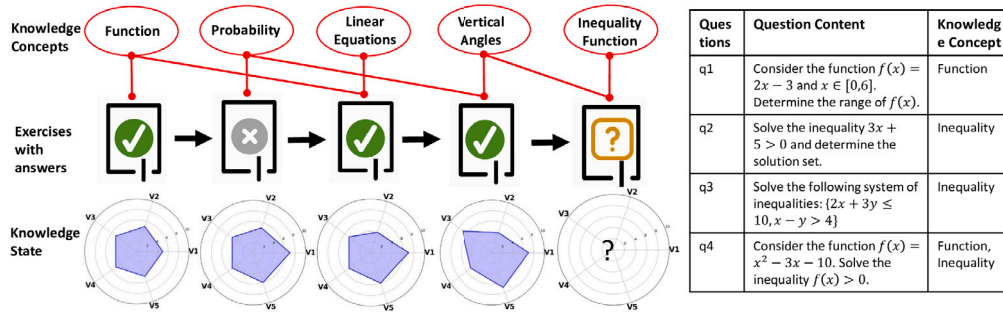


Fig. 1. Knowledge tracing framework linking exercises to knowledge concepts and student knowledge states.

we propose a Course Map and Difficulty enhanced Knowledge Tracing (CMDKT) framework, a novel approach that improves knowledge tracking by combining different educational data sources, concept map structure mining, and valuable question difficulty semantics to enhance the accuracy of predicting a learner's knowledge state. The primary contributions of our research can be summarized as follows:

- We analyze the significant effects of prerequisite relationships among concepts and the impact of question difficulty on students' learning.
- We introduce an innovative model called Course Map and Difficulty enhanced Knowledge Tracing (CMDKT) to enhance the prediction of students' knowledge state. In CMDKT, we utilize prerequisite relationships and difficulty levels to improve question representation. We also carefully designed a Gated Recurrent Unit (GRU) model to establish the connection between a student's knowledge state and the question embedding.
- Extensive experiments conducted on two public educational datasets confirm the effectiveness of CMDKT. Results indicate that the CMDKT model outperforms all baseline models by a minimum margin of 2.47% in terms of the average area under the Receiver Operating Characteristic (ROC) curve (AUC) and by 2.32% in overall prediction accuracy (ACC).

The organization of this paper is as follows: Section 2 provides a literature review that explores knowledge tracing. Section 3 presents the problem formulation and a detailed description of our model. Section 4 comprises a comprehensive set of experiments conducted to assess the effectiveness of our proposed method. Lastly, Section 5 summarizes the findings of our research and suggests potential directions for future studies.

2. Literature review

In the field of knowledge tracing, numerous efforts have been performed. The existing approaches can be categorized into the traditional and deep learning branches.

2.1. Traditional knowledge tracing

Corbett and Anderson were the pioneers in introducing the concept of knowledge tracing, using Bayesian networks to model the learning process. This approach, known as Bayesian Knowledge Tracing (BKT), marked the beginning of research in knowledge tracing [1]. BKT assumes that a student's knowledge state can be represented as a latent variable and uses Hidden Markov Model to update the knowledge state based on the observed student responses.

As the importance of knowledge tracing gained broader recognition, it began to attract increased attention from researchers. Various logistic models, including Learning Factor Analysis [4] and Performance Factor Analysis [2], were subsequently utilized for knowledge tracing. These

models aimed to unravel the underlying factors influencing learning progress. However, recent years have witnessed a paradigm shift in knowledge tracing research, driven by the transformative impact of deep learning techniques. The ability of deep learning to extract and represent complex features and unveil intricate structures has significantly propelled and advanced the field of knowledge tracing. We review some typical deep learning knowledge tracing models and the variations in the following subsections.

2.2. Deep knowledge tracing

A noteworthy advancement in the domain of deep learning knowledge tracing is the incorporation of recurrent neural networks (RNNs) within the Deep Knowledge Tracing model [3], which significantly improved performance over prior models. Building on this, researchers have explored various neural architectures to better capture student learning patterns.

One notable enhancement is the Dynamic Key-Value Memory Network (DKVMN), as introduced in the work by [5], which extends recurrent neural networks by integrating memory matrices to represent knowledge and learner mastery. However, DKVMN faces challenges in capturing long-term dependencies, prompting the development of the Sequential Key-Value Memory Network (SKVMN) [6]. SKVMN builds upon DKVMN's memory capacity while leveraging the recurrent modeling approach of Deep Knowledge Tracing (DKT). It employs a modified Long Short-Term Memory (LSTM) architecture called Hop-LSTM to better capture long-term dependencies, thereby allowing the model to consider the learner's evolving knowledge state for more nuanced predictions.

2.2.1. Attention-based knowledge tracing

In parallel, Attention-based Knowledge Tracing has emerged as a promising approach to overcome limitations inherent in traditional KT models, such as assumptions about learner knowledge states and the uniform relevance of all questions in a sequence. To overcome these challenges, researchers have incorporated attention mechanisms into KT models to dynamically learn the relevance of individual questions, as showcased in studies by [6–8]. These attention mechanisms enable models to assign varying degrees of importance to each question, thereby improving the accuracy and interpretability of predictions. The developments in both memory-augmented and attention-based KT methodologies demonstrate the ongoing efforts to enhance the predictive capabilities of KT models and their applicability across diverse learning contexts.

In addition, models like Exercise-Enhanced Recurrent Neural Network (EERNN) [9] leverage bi-directional LSTM to extract question representations. By utilizing both forward and backward LSTM layers, EERNN captures the temporal dependencies and contextual information present in the exercise sequences, enabling it to generate comprehensive question representations. While Exercise-Aware Knowledge Tracing (EKT) [10] incorporates text analysis to gauge question difficulty

and integrates multiple Knowledge Components (KCs). EKT leverages natural language processing (NLP) methods to analyze the textual content of questions, thereby capturing nuanced features related to question difficulty. Additionally, it considers the multidimensional nature of student knowledge by integrating diverse KCs. Relation-Aware Self-Attention for Knowledge Tracing (RKT) [11] combines attention weights and relation coefficients, utilizing question text information to estimate relationships in past interactions. It employs self-attention mechanisms to dynamically weigh the importance of each exercise in the sequence while considering the relational information between exercises. By incorporating question text information, it captures semantic similarities between exercises. Recent developments in knowledge diffusion modeling over graphs [12] and semantic-enhanced representation learning [13] also support the fusion of content features (e.g., question text, difficulty) with graph structures — a strategy adopted in CMDKT.

2.2.2. Graph-based knowledge tracing

Graph learning techniques have been widely applied to various knowledge representation tasks, including semantic link prediction and question answering. For example, Liu et al. [14] proposed a knowledge-based natural answer generation model via effective graph learning, which captures semantic relations in knowledge graphs to enhance response quality. Li et al. [13] introduced Hysae, an efficient semantic-enhanced representation learning model for knowledge hypergraph link prediction, demonstrating the advantages of hypergraph structures in modeling complex relational data. Zhang et al. [15] provided a comprehensive survey on temporal knowledge graph embedding methods, summarizing models and applications that capture temporal dynamics in knowledge graphs.

These studies highlight the value of leveraging structural and semantic information in graph-based models, which inspires our integration of course concept topology and question difficulty into the knowledge tracing process. A recent trend involves exploring graph representation learning techniques, particularly Graph Neural Networks (GNNs), for capturing relational structures in Knowledge Tracing tasks. Notable examples include Graph-based Knowledge Tracing (GKT) by Nakagawa et al. [16], treating knowledge tracing as a node-level classification problem using GNNs, and Graph-based Interaction Knowledge Tracing (GIKT) by Yang et al. [17], extending GKT by considering multiple relations between knowledge concepts (KCs) and questions. Another model, Structure-based Knowledge Tracing (SKT) [18], captures multiple relations among KCs and incorporates temporal and spatial effects for predicting student answers. These advancements collectively contribute to the evolving landscape of deep learning knowledge tracing methodologies. Ni et al. [19] introduced a model for tracking learners' short-term attentional knowledge through heterogeneous hierarchical differentiation. Utilizing hierarchical heterogeneous knowledge structures and short-term memory enhancement, they modeled the impact of various interaction sequences on learners. Specifically, it extracted knowledge structure features through a graph information augmentation component based on a heterogeneous graph framework. Additionally, it derived question differentiation parameters by adapting the TrueSkill system and emphasized learners' history-related practices through attention windowing. Further, Yang et al. [20] proposed a heterogeneous graph-based Knowledge tracing method. By tracing knowledge state evolution along temporal and spatial dimensions, it constructed a heterogeneous graph incorporating multiple exercise attributes. The method incorporated a spatial updating module to update node states within the knowledge space and propagate influence to neighboring nodes, alongside a temporal updating module for short-term memory enhancement and long-term memory forgetting.

This direction aligns with the broader trend in knowledge graph learning and representation, such as the work of Zedong Liu et al. [14], who proposed a knowledge-based natural answer generation method

using effective graph learning, and Zhao Li et al. [21], who developed HyCubE — a hypergraph convolutional embedding model, which can inform structure-aware KT modeling. These methods demonstrate the power of graph learning frameworks in capturing rich, structured relationships — a capability we leverage in CMDKT through concept maps.

2.2.3. Forgetting-aware knowledge tracing

In the realm of educational psychology, studies have underscored the importance of accounting for forgetting when accurately estimating a student's knowledge state, recognizing that mastery of knowledge tends to exhibit exponential decline over time since the last practice.

Forgetting-aware Knowledge Tracing models have emerged with the explicit goal of incorporating forgetting mechanisms into the estimation of student knowledge. An exemplar in this category is the DKTforget model [22], an extension of Deep Knowledge Tracing (DKT). DKTforget introduces three types of side information — repeated time gap, sequence time gap, and past trial count — to comprehensively model forgetting behavior. These additional features enhance the model's capacity to capture the nuances of forgetting, contributing to a more accurate estimation of a student's knowledge state.

In a parallel development, the HawkesKT model proposed by Wang et al. [23] incorporates the Hawkes process to model temporal cross-effects in Knowledge Tracing. The intensity function in HawkesKT captures the cumulative effects of prior learning interactions and their evolution over time, providing a sophisticated means of understanding the temporal dynamics of knowledge retention and decay.

As the field of knowledge tracing progresses, these forgetting-aware models deepen our understanding of how temporal factors influence learning outcomes. Beyond theoretical advancements, the practical application of knowledge tracing in real-world educational settings continues to evolve. This evolution contributes to the development of personalized learning systems and intelligent educational technologies, demonstrating the ongoing impact and relevance of knowledge tracing in diverse educational scenarios.

Our CMDKT model synthesizes insights from the above streams. It accounts for question difficulty, drawing inspiration from EKT [10] and knowledge-based QA approaches [24], while also incorporating temporal interaction patterns informed by the comprehensive survey on temporal knowledge graph embeddings by Yuchao Zhang et al. [15], as well as prior work on deep learning-based knowledge tracing [25] and contrastive learning approaches such as Bi-CLKT [26], which highlight the importance of modeling sequential and relational dependencies in student learning behavior.

3. Knowledge tracing based on course map

In this section, we introduce CMDKT, a knowledge tracing model that integrates the structural information from a course concept map with question difficulty to enhance prediction accuracy. Traditional KT models such as BKT, DKT, and GKT focus primarily on sequential interaction data and concept-question mappings, often overlooking the explicit prerequisite relationships between concepts or the varying difficulty levels of questions. By leveraging a course map, CMDKT encodes concept topology into the learner modeling process, enabling the model to capture how mastery of one concept can influence the acquisition of others. Additionally, incorporating question- and concept-level difficulty allows the model to differentiate between questions of varying cognitive demand. The following subsections define the KT task formally, outline the model's inputs and outputs, and present the architectural components of CMDKT.

Table 1
List of notation used in the model.

Notation	Description
S	Set of learners
Q	Set of unique questions
C	Set of knowledge concepts (KCs)
s_i	Series of question-answer pairs for learner u_i
N_i	Length of sequence for learner u_i
$q_{i,t}$	Question answered by learner u_i at time t
$a_{i,t}$	Correctness label at time t
X	Historical practice sequence
R	Question-concept matrix
G	Concept map
$x_{i,t}$	Input representing learner u_i 's history practice at t
e_k	Embedding of concept c_k
m_k	Hidden input of GRU in topology embedding
$y_{i,t}$	Knowledge state of learner u_i at time t

3.1. Problem definition

Let $S_i = \{(q_{i,1}, a_{i,1}), (q_{i,2}, a_{i,2}), \dots, (q_{i,T_i}, a_{i,T_i})\}$ denote the sequence of T_i interactions of learner i , where $q_{i,t}$ is the t th question attempted and $a_{i,t} \in \{0, 1\}$ indicates whether the learner answered correctly (1) or incorrectly (0). Each question q is associated with one or more concepts from a set $\mathcal{K} = \{1, 2, \dots, K\}$, as defined by a question-concept mapping matrix $R \in \{0, 1\}^{Q \times K}$.

The goal of the knowledge tracing (KT) task is to predict the learner's performance on the *next* question $q_{i,t+1}$ given their historical sequence $S_i^{(t)} = \{(q_{i,1}, a_{i,1}), \dots, (q_{i,t}, a_{i,t})\}$. We formalize this as a binary classification problem at the question level, estimating the probability $P(a_{i,t+1} = 1 \mid S_i^{(t)})$, which represents the likelihood that learner i will answer $q_{i,t+1}$ correctly.

Internally, CMDKT maintains a *concept-level mastery state* $y_{i,t,k} \in (0, 1)$ for each concept $k \in \mathcal{K}$. This quantity can be interpreted as a continuous-valued regression output representing the estimated probability that learner i has mastered concept k at time t . While $y_{i,t,k}$ itself is not directly supervised, it is used to compute $P(a_{i,t+1} = 1)$ through an aggregation function (see Eq. (11)) and is evaluated using both classification metrics (AUC, ACC) and regression-style metrics (MAE, RMSE) to assess probabilistic calibration.

Thus, CMDKT can be viewed as performing binary classification on next-question correctness, supported by a latent regression over per-concept mastery probabilities. A comprehensive list of key notations employed in this research is provided in Table 1 for ease of reference.

3.2. Model overview

CMDKT integrates *concept topology* and *question difficulty* into a unified deep knowledge tracing framework. Fig. 2 illustrates the architecture, which consists of three main modules:

1. **Topology Embedding Module** — encodes the course concept graph G into concept embeddings e_k that capture prerequisite relationships.
2. **Difficulty Encoding Module** — generates embeddings for both question-specific and concept-level difficulty, producing a combined difficulty vector d_p .
3. **Temporal Prediction Module** — a GRU-based recurrent network that processes the sequence of fused input vectors (question embedding, topology embedding, difficulty embedding, correctness encoding) to update the learner's knowledge state over time.

At each time step t , the model updates the learner's concept mastery state $y_{i,t,k}$ and computes the predicted probability $P(a_{i,t+1} = 1)$. The network is trained end-to-end using binary cross-entropy loss on the next-question correctness labels, optionally combined with an auxiliary MSE loss on $y_{i,t,k}$ to improve mastery calibration.

The following sections detail each module in turn.

Table 2
Prerequisite relationships between concepts in mathematics.

Concept	Prerequisite concepts
Addition Whole Numbers	Number Line
Mean	Addition Whole Numbers, Division Whole Numbers
Multiplication Fractions	Addition and Subtraction Fractions
Conversion of Fraction	Ordering Fractions,
Decimals to Percents	Ordering Positive Decimals
Pythagorean Theorem	Square Root

3.3. Concept map topology embedding module

To embed graph data into a low-dimensional space while preserving its topology, network embedding emerges as a natural and effective method. Network embedding techniques aim to map nodes in a graph to low-dimensional vectors while preserving the structural and relational information present in the original graph. These techniques have gained significant attention in recent years due to their ability to capture complex patterns and dependencies in graph-structured data. They have been successfully applied in various domains, including social network analysis, recommendation systems, and bioinformatics [27–29]. In our methodology, we begin by detailing the Prerequisite Relation Labeling and Validation process. Following that, we outline our modeling strategy, employing network embedding techniques to harness the topology information extracted from a concept map, thus facilitating the understanding of its structural characteristics.

3.3.1. Prerequisite relation labeling and validation

Understanding the prerequisite relationships between concepts is essential for designing effective learning pathways in mathematics education. A multi-stage approach was employed to establish prerequisite relationships between concepts in datasets from the mathematics and database systems domains, as shown in prior work [30–32]. Initially, the prerequisite relationships were generated using the ChatGPT 4 language model, which leverages state-of-the-art natural language processing techniques to analyze and infer relationships from educational texts and materials [33–35]. Subsequently, two experts specializing in mathematics education and another two experts in database systems were engaged to review and validate the generated relationships. Their expertise in the respective domains ensured a thorough assessment of the logical coherence and educational validity of the prerequisite dependencies identified by the language model.

The validation process involved a detailed examination of each relationship to ensure it accurately reflected the logical progression of learning in both mathematics and database systems areas. Any discrepancies or ambiguities in the generated relationships were collaboratively identified and resolved through rigorous discussion and deliberation between the experts. The engagement of subject matter experts not only served to validate the generated relationships but also contributed to the overall robustness and credibility of the findings. Their domain knowledge ensured that the prerequisite relationships were educationally sound and relevant to both mathematics and database systems education.

Table 2 illustrates selected examples derived from the generated concept map, showcasing the prerequisite relationships between various mathematical concepts. For instance, mastery of the Pythagorean theorem requires a solid understanding of square roots, which form the basis for calculating the lengths of sides in a right triangle. Similarly, proficiency in multiplication of fractions necessitates a strong foundation in addition and subtraction of fractions, as multiplication often involves repeated addition or subtraction of fractional quantities. These prerequisite relationships serve as building blocks for students' mathematical understanding, allowing educators to design curricula that scaffold knowledge acquisition and facilitate conceptual understanding.

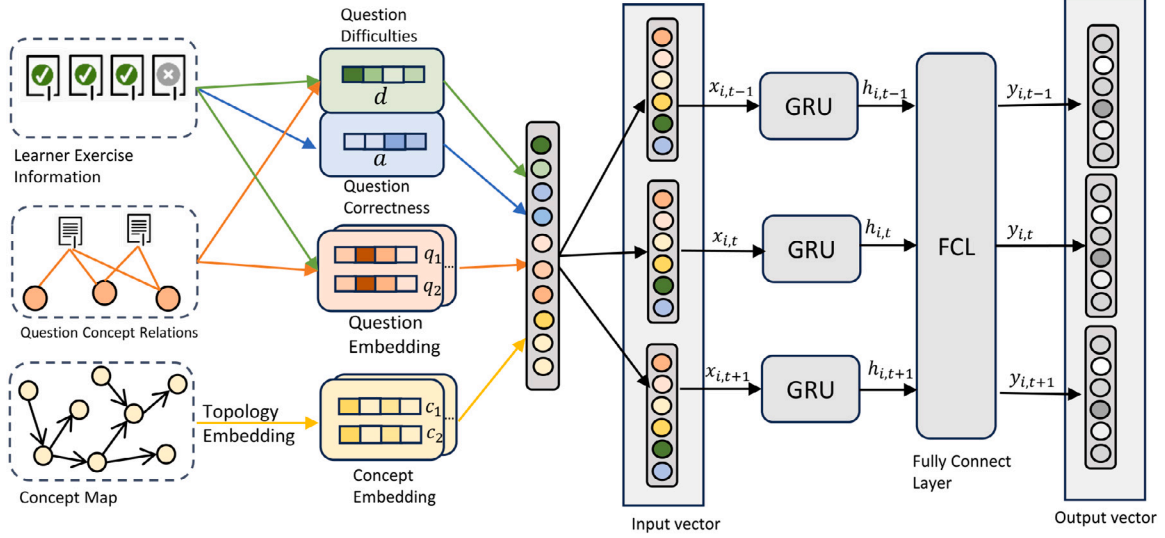


Fig. 2. Course Map and Difficulty enhanced Knowledge Tracing Model.

Ultimately, the collaborative efforts of leveraging advanced language models and engaging domain experts yielded a comprehensive and validated set of prerequisite relationships between concepts in mathematics and database systems. These relationships form the foundation for further analysis and interpretation, facilitating more effective curriculum design, instructional planning, and educational decision-making in both domains.

3.3.2. Concept map structure modeling through GRU

The Gated Recurrent Unit (GRU) [36] effectively models sequences of variable length, which is essential for processing concept maps that may vary in size and complexity. Compared to other recurrent neural network (RNN) architectures such as the vanilla RNN [37] or LSTM (Long Short-Term Memory) [38], GRUs have fewer parameters, making them computationally more efficient while maintaining expressive power. By leveraging these properties of GRUs, we can enhance the learning process and effectively capture the rich structural and dependency information inherent in concept maps. Concepts in educational domains often have hierarchical dependencies, where understanding one concept relies on grasping prerequisite concepts. Therefore the prerequisites of a concept provide valuable semantic information about the concept itself. Expanding upon the standard GRU, which typically includes a reset gate, an update gate, a temporary hidden state, and an output hidden state, our proposed model takes a distinct approach to computing inputs for these gates. To consider the dependencies and relationships between the current concept and its prerequisite concepts, the unit integrates two types of input information: (1) The feature vector v_k , which represents the pre-trained representation of concept c_k , captures its intrinsic characteristics and properties. (2) The hidden input m_k , influenced by the concept's prerequisites, encapsulates the contextual information and dependency relationships associated with concept c_k .

As depicted in Fig. 3, Concept 1 is indicated as a prerequisite concept of both Concept 2 and Concept 3. Therefore, the embedding of Concept 1, denoted as e_1 , is fed into the GRU as the hidden input for calculating the embeddings e_2 and e_3 . This allows the model to consider the dependencies and relationships between the current concept and its prerequisite concepts, leading to more informed concept embeddings.

With the defined feature vector v_k and the hidden input m_k , the proposed GRU network further applies a series of operations similar to the standard GRU to derive the concept embedding e_k , which can be summarized as follows:

$$r_k = \sigma(W_r v_k + U_r m_k + b_r) \quad (1)$$

$$z_k = \sigma(W_z v_k + U_z m_k + b_z) \quad (2)$$

$$\tilde{e}_k = \tanh(W_e v_k + U_e [r_k \odot m_k] + b_e) \quad (3)$$

$$e_k = (1 - z_k) \odot m_k + z_k \odot \tilde{e}_k, \quad (4)$$

where $Q_c = \{W_r, U_r, b_r, W_z, U_z, b_z, W_e, U_e, b_e\}$ are the internal parameters and $\odot$ denotes the pointwise multiplication. The obtained concept embedding e_k essentially incorporates the concept prerequisite dependency information, where the structure of the GRU network explicitly follows the topology of the concept map. Additionally, this GRU network can be co-trained with the other parts of the model.

3.4. Question difficulty encoding module

The connection between the difficulty levels of questions and students' knowledge states serves as a natural indicator for estimating their proficiency levels. Students who can successfully answer more challenging questions are typically considered to possess higher levels of knowledge. This association between question difficulty and students' knowledge states has been validated by numerous previous studies. In psychological theories like the Classic Test Theory (CTT) [39], question difficulty is often quantified statistically, representing the proportion of a defined group of students who answered a question correctly. Similarly, the Item Response Theory (IRT) [40] posits that question difficulty and student ability are probabilistically dependent on students' responses, typically measured using logistic functions. Moreover, the difficulty of a question correlates positively with the number of knowledge concepts (KCs) it encompasses [41]. This study aims to enhance Knowledge Tracing (KT) performance by clarifying the relationship between students' knowledge states and the difficulty levels of questions.

3.4.1. Difficulty definition

As the Knowledge Concepts (KCs) contained in a question also have distinctive difficulties (e.g., Multiplication and Division of Fractions are harder than Addition and Subtraction of Fractions), we define the question difficulty in two levels: Question Specific (QS) difficulty and Knowledge Concept (KC) difficulty. Inspired by Classical Test Theory (CTT) and some previous works [39,42,43], we use an objective statistical means to calculate the QS and KC difficulty as follows:

$$QS = \frac{\sum_i |S_i| \cdot \{a_{ij} = 1\}}{\sum_i |S_i|} \cdot C_{qs} \quad (5)$$

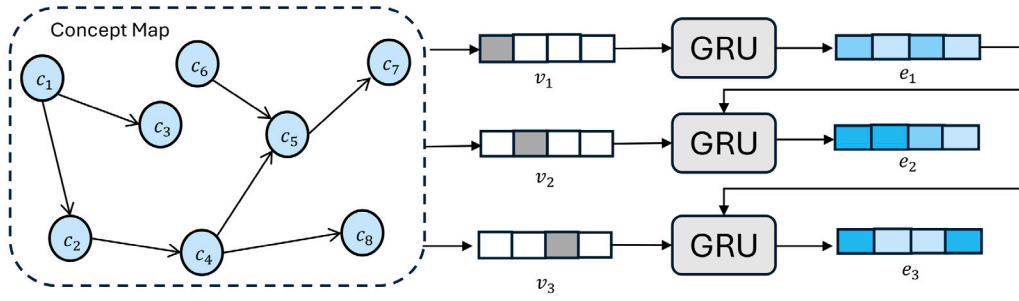


Fig. 3. Concept structure information Embedding through GRU.

$$KC = \frac{\sum_i |S_i| \cdot \{a_{im} = 1\}}{\sum_i |S_i|} \cdot C_{kc} \quad (6)$$

where S_i is the set of students who answered the question q_j or the knowledge concept c_m , $a_{ij} = 1$ and $a_{im} = 1$ refer to corresponding correct answers respectively, and the constants C_{qs} and C_{kc} are the predefined levels of QS and KC difficulty.

3.4.2. Enhanced difficulty-aware question embedding

Difficulty embedding is the embedding of the question with difficulty aiming to enrich question representations. Recent advancements in Knowledge Tracing (KT) models have recognized the significance of incorporating question difficulty within question embeddings, as evidenced by several studies [42,44,45]. However, these models typically leverage question difficulty information implicitly and they often overlook the relative difficulty levels of Knowledge Concepts (KCs), which are essential supplementary details for Question Specific (QS) difficulty estimation. In our approach, known as CMDKT, we capitalize on both QS and KC difficulty metrics to refine the question representation. Since question difficulty encompasses both QS and KC difficulty, the difficulty embedding is composed of both QS and KC embeddings. More specifically, as we have defined C_{qs} different difficulty levels for QS , we represent the QS embedding by an embedding matrix $Q_S \in R^{C_{qs} \times d_{qs}}$, where d_{qs} is the dimension. Therefore, for a question q_j with a specific difficulty level C_{jq_s} of QS , we can directly obtain its QS embedding from Q_S . Similarly, we use an embedding matrix $K_C \in R^{C_{kc} \times d_{kc}}$ to represent the KC embedding. We can also obtain the KC embedding of a question according to its difficulty level of KC . We integrate the original question embedding $q_{i,t}$ with QS difficulty Q_S and KC difficulty K_C , respectively to generate the enhanced question embedding $x_{i,t}$.

3.5. Temporal prediction module

The temporal prediction module is responsible for modeling the evolution of a learner's knowledge state over time. At each interaction step t , the model receives a fused representation of the current question, the relevant concept topology, the associated difficulty information, and the learner's response correctness. This representation is passed through a gated recurrent unit (GRU) to update the hidden state, which encodes the learner's dynamic knowledge state. From this hidden state, the model computes the per-concept mastery probabilities $y_{i,t,k}$ and subsequently the probability of correctly answering the next question $P(a_{i,t+1} = 1)$.

As depicted in Fig. 2, the CMDKT model incorporates three primary inputs: learner exercise data, question-concept relations, and the topological details of the concept map. For a given learner u_i , their exercise sequence s_i comprises question-answer pairs. Each question is then represented by a dense vector $q_{p(i,t)}$ of dimension j_Q via an embedding layer. The answer information is encoded as a zero vector of the same dimension j_Q , following the approach in prior research [10].

Additionally, the CMDKT model leverages the prerequisite dependency information derived from the concept map structure, which is captured through a dedicated topology embedding module (discussed in Section 3.3). Furthermore, the difficulty levels associated with questions are obtained via an embedding layer that incorporates question and concept difficulty factors (detailed in Section 3.4).

Therefore, the CMDKT model input is composed of four components as follows:

$$x_{i,t} = \begin{cases} q_{p(i,t)} \oplus e_{p(i,t)} \oplus d_{p(i,t)} \oplus \mathbf{0}, & \text{if } a_{i,t} = 1 \\ \mathbf{0} \oplus q_{p(i,t)} \oplus e_{p(i,t)} \oplus d_{p(i,t)}, & \text{otherwise} \end{cases} \quad (7)$$

$\oplus$ denotes the concatenation operation, $q_{p(i,t)}$ is the dense vector of each question, which is the output of an embedding layer; $e_{p(i,t)}$ represents the concept embedding for question $q_{p(i,t)}$, computed by aggregating the embedding result of the first GRU of all concepts contained in question $q_{p(i,t)}$ through pooling:

$$e_{p(i,t)} = \text{Pool}(\{e_{c_k} | c_k \in q_{p(i,t)}\}) \quad (8)$$

Where $c_k \in q_{p(i,t)}$ represents the concept c_k belongs to question $q_{p(i,t)}$, and the e_{c_k} is the embedding of concept c_k from topology embedding module. Moreover, $d_{p(i,t)}$ is the difficulty embedding result utilizing the question difficulties and concept difficulties. $a_{i,t}$ is a binary representation where 1 indicates that learner u_i correctly answered question $q_{p(i,t)}$ at time t . We use the position of $\mathbf{0}$ vector with the same dimension as $q_{p(i,t)}$ to distinguish between correct and incorrect answers.

Our CMDKT model is designed to evaluate the knowledge mastery state of learner u_i at time t on concept c_k , represented as $y_{i,t,k}$. It utilizes a standard GRU module to process the input $x_{i,t}$, resulting in the output $h_{i,t}$ at time t . The GRU operation is defined as:

$$h_{i,t} = f(h_{i,t-1}, x_{i,t}) \quad (9)$$

where $h_{i,t}$ and $h_{i,t-1}$ denote the outputs of the GRU model at time t and $t-1$ respectively. The activation function $f(\cdot)$ follows the standard GRU implementation, with its internal parameters represented by Q_S .

Given $y_{i,t,k}$ as the knowledge state of learner u_i on concept c_k at time t , the likelihood of the learner mastering that concept can be calculated as:

$$P(y_{i,t,k} = 1) = \sigma(W_y h_{i,t} + b_y)^T c_k \quad (10)$$

Here, c_k represents the embedding of concept k , W_y and b_y are the parameters of the fully connected layer, and $\sigma(\cdot)$ means the sigmoid function. We treat $y_{i,t,k} \in (0, 1)$ as the model's estimate of the probability that learner i has mastered concept k at time t .

For evaluation, we convert mastery probabilities into next-question correctness predictions using Eq. (11). For ACC, we threshold $P(a_{i,t} = 1)$ at 0.5 to obtain binary predictions, while AUC is computed directly from the probabilistic outputs. The model is trained primarily with binary cross-entropy (BCE) loss on the observed next-question correctness labels $a_{i,t}$. Optionally, an auxiliary mean squared error (MSE) loss on $y_{i,t,k}$ can be included to regularize concept-level mastery predictions.

$$P(a_{i,t} = 1) = \prod_{c_k \in q_{p(i,t)}} P(y_{i,t-1,k} = 1) \quad (11)$$

Table 3
Statistics of datasets.

Statistics	ASSIST09	DBE-KT22
# of practice records	172,608	159,695
# of students	1,712	1,034
# of questions	1,084	212
# of KCs	68	98
Avg. records per student	100.82	154.44
Avg. records per question	159.23	753.28

where $c_k \in q_{p(i,t)}$ denotes the question $q_{p(i,t)}$ that contains concept c_k . In Eq. (11), the per-concept mastery probabilities are multiplied, implying conditional independence of concepts given the student's state. Based on these mastery probabilities, the model estimates the likelihood that a learner will correctly answer the next question, which depends on their proficiency in all relevant concepts associated with that question.

This formulation allows CMDKT to jointly capture the temporal dynamics of learning, the structural relationships between concepts, and the varying difficulty of questions. By maintaining a concept-level mastery state $y_{i,t,k}$ and using it to compute question-level correctness probabilities, the model bridges fine-grained knowledge representation with the practical objective of predicting learner performance. The combination of topology-aware embeddings, difficulty encoding, and recurrent temporal modeling enables CMDKT to adapt to diverse learning trajectories, providing both interpretable mastery estimates and accurate next-question predictions.

4. Experiments

In this section, we present the experimental setup, which encompasses the real-world datasets, baseline models, and settings. Following this, we evaluate the proposed CMDKT model and conduct comparative analyses against the baseline models. Additionally, we conducted several experiments aimed at demonstrating interpretability by analyzing the influence of prerequisite numbers and visualizing the Process of Predicting Students' Performance.

4.1. Datasets and experimental setup

For our experiments, we chose two real-world datasets to assess the efficacy of CMDKT, each offering unique insights into student learning behavior. The first dataset, ASSIST2009-2010 (ASSIST09) is a widely used dataset in Knowledge Tracing and educational data mining research, which is collected from the ASSISTments online tutoring platform [46]. We utilized the “non-skill builder” dataset and performed specific filtering operations for preprocessing. First, we selected sequences generated with “random order” to mitigate potential bias from imposed order. Subsequently, we filtered sequences longer than 30 items to ensure an adequate amount of learner information. After preprocessing, ASSIST09 comprises data from 1712 students tackling 1084 questions across 68 skills, resulting in a total of 172,608 records.

The second dataset, DBE-KT22 (Database Exercises for Knowledge Tracing) [47], originates from a course titled “Introduction to Relational Databases” from IT area at the Australian National University (ANU). Spanning academic years 2018 to 2021, this dataset encompasses interactions from 1361 students across various disciplines such as computer science, engineering, arts, and business. We also filtered the students with sequences longer than thirty, then it features 212 questions covering 98 distinct concepts, amounting to a comprehensive dataset of 159,695 records from 1034 students. The statistics of the datasets are listed in Table 3.

In our experiments, we standardized all exercise records by sorting them according to the timestamp of answering. Optimization of our model was carried out using the Adam algorithm, with a mini-batch size of 128 for both datasets and a learning rate set to 0.001.

The hidden dimensions for the topology embedding GRU and the subsequent knowledge state prediction GRU are set to 32. Standard 5-fold cross-validation was employed for all models, with 90% of the learning sequences split as the training set and validation set in each fold (in a 9:1 ratio), and the remaining 10% used as the testing set. Hyperparameters were learned exclusively on the training set, and the model that performed best on the validation set was utilized for evaluating the testing set. All models were implemented using PyTorch in Python on a Windows server equipped with 32 GB of RAM, an Intel i7 CPU, and one GeForce 4090 GPU.

4.2. Baselines

For this study, we employ three established knowledge tracing (KT) models as our baseline benchmarks: Bayesian Knowledge Tracing (BKT), Deep Knowledge Tracing (DKT), and Graph Knowledge Tracing (GKT). We also include the CMDKT model without difficulty module (named as CMDKT/D) for the ablation study.

Bayesian Knowledge Tracing (BKT) [1], a foundational and widely adopted model in knowledge tracing tasks, uses a binary variable to represent whether a concept has been mastered. It applies a hidden Markov model (HMM) to iteratively update the probability of mastery based on the sequence of observed learner responses. BKT accounts for factors like guessing and slipping while assuming that once a concept is mastered, it is not forgotten.

Deep Knowledge Tracing (DKT) [3] utilizes deep learning techniques and recurrent neural network (RNN) structures with Long Short-Term Memory (LSTM) units to model the learning process. It offers a continuous and intricate representation of the learner's knowledge state, thereby enhancing its ability to capture complex learning dynamics. CMDKT, a variant of DKT, incorporates educational relation information and topological information from the concept map while employing a similar RNN structure.

Graph Knowledge Tracing (GKT) [16] employs a graph neural network (GNN) model to track learners' evolving knowledge states. By leveraging the provided knowledge structure, GKT computes the updated mastery state of a concept by aggregating the states of its neighboring concepts. It iteratively updates the neighboring concepts based on the computed state, ultimately estimating the learner's knowledge states across all concepts. This approach allows GKT to model the dynamic learning process by leveraging the connectivity and dependencies inherent in the underlying knowledge structure.

CMDKT/D denotes the CMDKT model excluding the consideration of question difficulties. We leverage the concept map configuration to generate concept embeddings, akin to the CMDKT model, and employ a secondary GRU to forecast the mastery state of knowledge as a reference point. This baseline aims to offer insights into the efficacy of both the concept prerequisite topology and question difficulty components.

4.3. Evaluation

To evaluate the overall effectiveness of our model, we focus on predicting students' future performance by assessing the correctness of the following questions, where higher accuracy implies a more accurate estimate of their knowledge state. We compare CMDKT against all baseline models. Our task involves predicting student performance at each timestep t based on their performance from timestep 1 to $t - 1$.

To ensure comprehensive evaluation, we employ four key metrics across all experiments: mean absolute error (MAE) and root Mean Squared Error (RMSE) are used to evaluate our model considering it is a regression problem, the overall prediction accuracy (ACC) and the area under ROC curve (AUC) are used for evaluation based on the classification setting. This approach aims to provide robust evaluation results and insights into the predictive capabilities of CMDKT compared to baseline models. Employing an open-source platform, we implement

Table 4
Comparison of model performance on ASSIST09 and DBE-KT22 datasets.

Models	ASSIST09				DBE_KT22			
	auc	acc	mae	rmse	auc	acc	mae	rmse
BKT	0.7048	0.6482	0.4341	0.4658	0.7582	0.76	0.3918	0.4219
DKT	0.7857	0.7491	0.3892	0.4214	0.7994	0.7945	0.3504	0.3919
GKT	0.7524	0.733	0.4156	0.4565	0.7673	0.7534	0.3768	0.4112
CMDKT/D	0.8044	0.7625	0.3209	0.4	0.8016	0.7958	0.2987	0.3826
CMDKT	0.8051	0.7665	0.3157	0.4001	0.8088	0.7976	0.2756	0.3733

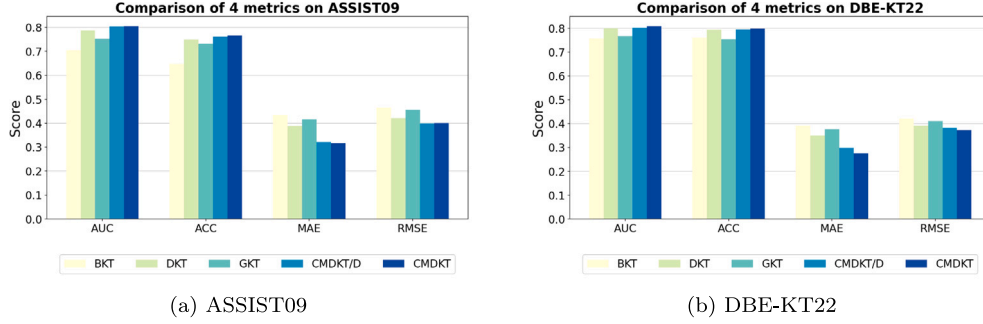


Fig. 4. Experimental results compared to the baselines.

the majority of the comparative baseline models and conduct a hyperparameter search to optimize their performance individually. Across all models, we conduct five separate experiments, each involving testing various parameters. Subsequently, we calculate the average metrics from these experiments to serve as evaluation metrics.

We evaluated the performance of CMDKT against several baseline models, namely BKT, DKT, GKT, and CMDKT/D. Notably, the CMDKT/D model refers to our CMDKT model without the incorporation of difficulty levels. The objective was to gauge the effectiveness of CMDKT in improving knowledge-tracing outcomes across different educational contexts. Table 4 displays the prediction performance of all compared models. The proposed CMDKT model achieves the highest AUC and ACC across two datasets, and the lowest MAE and RMSE, though the RMSE on the ASSIST09 dataset is nearly the same with CMDKT/D. For instance, in the ASSIST09 dataset, CMDKT achieved an impressive Area Under the Curve (AUC) score of 0.8051, representing a notable improvement over the other models. Similarly, on the DBE-KT22 dataset, CMDKT achieved an AUC of 0.8088, showcasing its robust performance across diverse educational domains. In addition to AUC, CMDKT also demonstrated superior accuracy (ACC) compared to the baselines. On the ASSIST09 dataset, CMDKT achieved an average accuracy of 0.7665, while on the DBE-KT22 dataset, it achieved an even higher ACC of 0.7976. These results signify a substantial enhancement in prediction accuracy, highlighting the efficacy of CMDKT in accurately forecasting student performance. Furthermore, CMDKT exhibited significant reductions in Mean Absolute Error (MAE) and Root Mean Square Error (RMSE) compared to the baseline models. On both datasets, CMDKT consistently achieved lower MAE and RMSE values, indicating its superior predictive accuracy and robustness in capturing student learning behavior.

Furthermore, CMDKT outperforms CMDKT/D across key metrics, highlighting the effectiveness of incorporating question difficulties into the prediction of students' knowledge states. Additionally, the superior performance of CMDKT/D over other baseline models underscores the significance of considering prerequisite dependencies when assessing knowledge states. The comparative analysis presented in Fig. 4 also provides comprehensive insights into the performance of CMDKT relative to other baseline models on both datasets. The results vividly illustrate the superior performance of CMDKT across multiple evaluation metrics.

Overall, our evaluation demonstrates the remarkable improvement achieved by CMDKT over existing baseline models in knowledge tracing

tasks. Across both the ASSIST09 and DBE-KT22 datasets, CMDKT consistently delivers superior performance, offering a promising avenue for enhancing educational analytics and personalized learning experiences. Moreover, these improvements represent a substantial percentage increase over the baseline models, underscoring the efficacy of CMDKT in leveraging the rich information encapsulated within concept maps and question difficulties to refine knowledge state estimation and enhance prediction accuracy.

4.4. Impact of the prerequisite number

We also examine the influence of varying the percentage of prerequisite pairs utilized in the model (ranging from 20% to 100%) on the four performance metrics. As depicted in Fig. 5, the experimental findings reveal a consistent improvement in the performance of CMDKT and CMDKT-D with an increasing number of prerequisite pairs considered, as accuracy (ACC) and AUC increases steadily as the percentage of prerequisite pairs rises from 20% to 100%. Specifically, the accuracy values are as follows: 0.7845 at 20%, 0.7877 at 40%, 0.7920 at 60%, 0.7951 at 80%, and 0.7976 at 100% on the DBE-KT22 dataset. The same trend is observed on the ASSIST dataset as well. This trend underscores the significance and effectiveness of integrating educational relationships into the KT model. Furthermore, across all percentages, the CMDKT model consistently demonstrates superior performance compared to the CMDKT-D model on each dataset. This suggests the effectiveness of integrating difficulty levels into our KT model for enhanced performance.

4.5. Case study: Visualizing the process of predicting students' performance

We illustrate the predictive capabilities of CMDKT using two students from the DBE-KT22 dataset. As depicted in Fig. 6, CMDKT demonstrates remarkable interpretability, providing not only accurate forecasts but also insights into the reasons behind student performance. For instance, consider student *S1*, who initially answered a question incorrectly, characterized by KC 92: Entity-Relationship Model and *QS* difficulty levels of 0.39 and 0.6. CMDKT accurately predicts that *S1* will answer the same question incorrectly when the *QS* difficulty levels increase to 0.73 and 0.63, indicating a harder question. Moreover, despite *S1* incorrectly answering a question with KC 117: Functional dependencies and *QS* difficulty levels of 0.61 and 0.69,

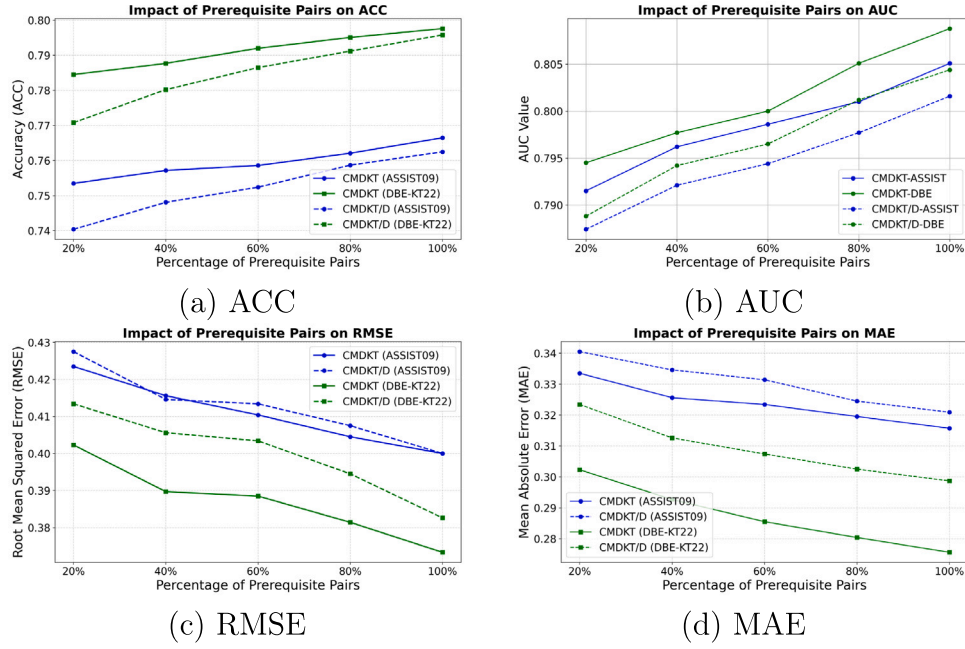


Fig. 5. The influence of the number of prerequisite pairs is illustrated, with distinct colors denoting various datasets. Solid lines correspond to CMDKT models, while dotted lines represent CMKT/D models. ASSIST and DBE represent the ASSIST09 and DBE-KT22 datasets, respectively. Panels (a) through (d) depict ACC, AUC, RMSE and MAE metrics.

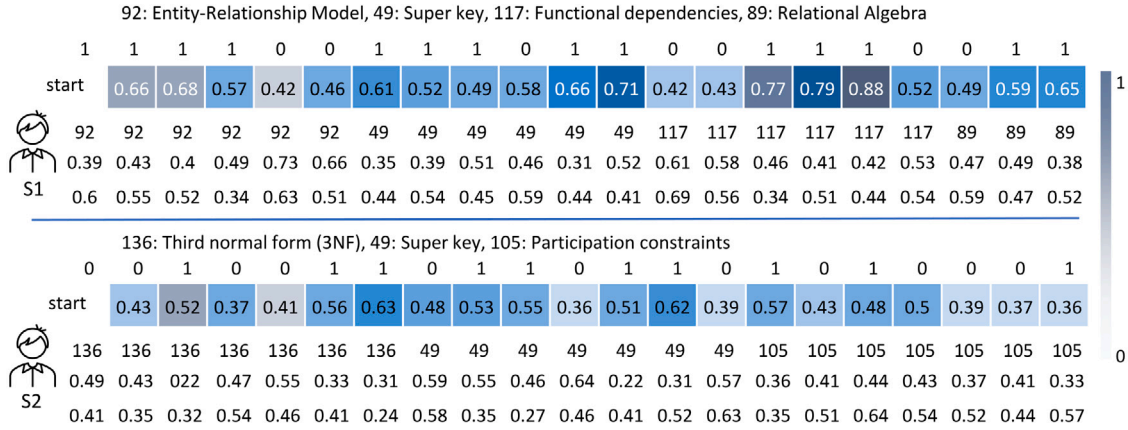


Fig. 6. Learning sequences and performance predictions of two students across different questions with varying difficulty levels in the DBE-KT22 dataset.

CMDKT forecasts $S1$'s success on a similar question with lower Q S difficulty levels of 0.46 and 0.34. Similarly, for student $S2$, CMDKT's predictions exhibit interpretability on KCs 136 (Third normal form) and 49 (Super key). However, $S2$'s performance on KC 105: Participation constraints fluctuate significantly, and CMDKT's forecasts often deviate from the actual answers. This suggests that $S2$ may display irregular learning behaviors, such as guessing or carelessness. Overall, CMDKT's predictions align with our expectations, indicating a higher likelihood of correct answers for questions with lower difficulty levels.

5. Conclusion

In this paper, we proposed CMDKT, a knowledge tracing model that integrates course concept topology and question difficulty into a unified framework. Experimental results on the ASSIST09 and DBE-KT22 datasets demonstrate that CMDKT outperforms strong baselines such as DKT, BKT, and GKT in terms of AUC, ACC, MAE, and RMSE, confirming the effectiveness of incorporating structural and difficulty information in predicting learner performance.

Limitations and future directions. CMDKT improves prediction by exploiting concept topology and difficulty, but several limitations remain. (1) Equation (11) multiplies per-concept mastery probabilities, which assumes independence across concepts; future work should explore learned fusion (e.g., attention or logistic models) to account for concept interactions. (2) We do not explicitly model time-dependent forgetting; incorporating time-gap features (as in DKTforget or HawkesKT) could capture decay effects. (3) The current $V_{i,t,k}$ is a unidimensional mastery probability per concept — extending CMDKT to model finer-grained knowledge mastery could enable richer learner profiling. For example, each concept could be represented by multiple skill dimensions (e.g., procedural skill, conceptual understanding, problem-solving strategy), with partial-credit scoring or graded mastery levels instead of a binary framework. (4) We evaluated on ASSIST09 and DBE-KT22; broader cross-domain validation and release of code and seeds would strengthen reproducibility. Finally, reporting model training time and complexity will help practitioners assess deployment cost.

We plan to pursue these directions in future work, with a particular focus on developing multi-dimensional mastery modeling, time-aware

learning, and more flexible concept interaction mechanisms to further advance the accuracy and interpretability of knowledge tracing models.

CRedit authorship contribution statement

Jianing Xia: Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Conceptualization. **Amanda Li:** Writing – review & editing. **Hui Yin:** Writing – review & editing. **Guozhe Jin:** Writing – review & editing, Validation, Methodology, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author(s) used ChatGPT to assist with paraphrasing and language refinement. After using this tool, the author(s) reviewed and edited the content as necessary and take full responsibility for the integrity and accuracy of the final manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

- Corbett AT, Anderson JR. Knowledge tracing: Modeling the acquisition of procedural knowledge. *User Model User-Adapt Interact* 1994;4:253–78.
- Pavlik PI, Cen H, Koedinger KR. Performance factors analysis - A new alternative to knowledge tracing. In: Dimitrova V, Mizoguchi R, du Boulay B, Graesser AC, editors. *Artificial intelligence in education. Frontiers in artificial intelligence and applications*, vol. 200, IOS Press; 2009, p. 531–8.
- Piech C, Bassen J, Huang J, Ganguli S, Sahami M, Guibas LJ, Sohl-Dickstein J. Deep knowledge tracing. In: Cortes C, Lawrence ND, Lee DD, Sugiyama M, Garnett R, editors. *Advances in neural information processing systems* 28. 2015, p. 505–13.
- Cen H, Koedinger K, Junker B. Learning factors analysis – a general method for cognitive model evaluation and improvement, vol. 4053. 2006, p. 164–75. http://dx.doi.org/10.1007/11774303_17.
- Zhang J, Shi X, King I, Yeung D. Dynamic key-value memory networks for knowledge tracing. In: *Proceedings of the 26th international conference on world wide web*. ACM; 2017, p. 765–74.
- Abdelrahman G, Wang Q. Knowledge tracing with sequential key-value memory networks. In: *Proceedings of the 42nd international ACM SIGIR conference on research and development in information retrieval*. 2019, p. 175–84.
- Choi Y, Lee Y, Cho J, Baek J, Kim B, Cha Y, Shin D, Bae C, Heo J. Towards an appropriate query, key, and value computation for knowledge tracing. In: *L@S'20: seventh ACM conference on learning @ scale*, virtual event. ACM; 2020, p. 341–4.
- Ghosh A, Heffernan NT, Lan AS. Context-aware attentive knowledge tracing. In: *KDD '20: the 26th ACM SIGKDD conference on knowledge discovery and data mining*, virtual event. ACM; 2020, p. 2330–9.
- Su Y, Liu Q, Liu Q, Huang Z, Yin Y, Chen E, Ding CHQ, Wei S, Hu G. Exercise-enhanced sequential modeling for student performance prediction. In: *Proceedings of the thirty-second AAAI conference on artificial intelligence*. AAAI Press; 2018, p. 2435–43.
- Liu Q, Huang Z, Yin Y, Chen E, Xiong H, Su Y, Hu G. EKT: exercise-aware knowledge tracing for student performance prediction. *IEEE Trans Knowl Data Eng* 2021;33(1):100–15.
- Pandey S, Srivastava J. RKT: relation-aware self-attention for knowledge tracing. In: *CIKM '20: the 29th ACM international conference on information and knowledge management*, virtual event. ACM; 2020, p. 1205–14.
- Xie T, Chen Y, Wu Y, Cui N, Chen H, Lu X, Huang X, Wang Y, Li J. Trust-based core social graph convolution: An innovative framework for location recommendation. *Expert Syst Appl* 2025;285:127899.
- Li Z, Wang X, Zhao J, Feng F, Chen Z, Li J. HySAE: An efficient semantic-enhanced representation learning model for knowledge hypergraph link prediction. In: Long G, Blumstein M, Chang Y, Lewin-Eytan L, Huang ZH, Yom-Tov E, editors. *Proceedings of the ACM on web conference 2025, WWW 2025*. ACM; 2025, p. 86–97. <http://dx.doi.org/10.1145/3696410.3714549>.
- Liu Z, Li J, Huang Y, Cui N, Pei L. Knowledge-based natural answer generation via effective graph learning. *Knowl-Based Syst* 2025;316:113288.
- Zhang Y, Kong X, Shen Z, Li J, Yi Q, Shen G, Dong B. A survey on temporal knowledge graph embedding: Models and applications. *Knowl-Based Syst* 2024;304:112454.
- Nakagawa H, Iwasawa Y, Matsuo Y. Graph-based knowledge tracing: Modeling student proficiency using graph neural networks. *Web Intell* 2021;19(1–2):87–102.
- Yang Y, Shen J, Qu Y, Liu Y, Wang K, Zhu Y, Zhang W, Yu Y. GIKT: a graph-based interaction model for knowledge tracing. In: *Machine learning and knowledge discovery in databases - European conference, ECML PKDD 2020. Lecture notes in computer science*, vol. 12457, Springer; 2020, p. 299–315.
- Tong S, Liu Q, Huang W, Huang Z, Chen E, Liu C, Ma H, Wang S. Structure-based knowledge tracing: An influence propagation view. In: *20th IEEE international conference on data mining*. IEEE; 2020, p. 541–50.
- Ni Q, Wei T, Zhao J, He L, Zheng C. HHSKT: A learner-question interactions based heterogeneous graph neural network model for knowledge tracing. *Expert Syst Appl* 2023;215:119334.
- Yang H, Hu S, Geng J, Huang T, Hu J, Zhang H, Zhu Q. Heterogeneous graph-based knowledge tracing with spatiotemporal evolution. *Expert Syst Appl* 2024;238(Part F):122249. <http://dx.doi.org/10.1016/j.eswa.2023.122249>.
- Li Z, Wang X, Zhao J, Guo W, Li J. HyCubE: Efficient knowledge hypergraph 3D circular convolutional embedding. *IEEE Trans Knowl Data Eng* 2025;37(4):1902–14.
- Nagatani K, Zhang Q, Sato M, Chen Y, Chen F, Ohkuma T. Augmenting knowledge tracing by considering forgetting behavior. In: *The world wide web conference*. ACM; 2019, p. 3101–7.
- Wang C, Ma W, Zhang M, Lv C, Wan F, Lin H, Tang T, Liu Y, Ma S. Temporal cross-effects in knowledge tracing. In: *WSDM '21, the fourteenth ACM international conference on web search and data mining*, virtual event. Israel, March 8–12, 2021, ACM; 2021.
- Fan Y, Ren T, Huang C, Zheng B, Jing Y, He Z, Li J, Li J. A confidence-based knowledge integration framework for cross-domain table question answering. *Knowl-Based Syst* 2024;306:112718.
- Song X, Li J, Cai T, Yang S, Yang T, Liu C. A survey on deep learning based knowledge tracing. *Knowl-Based Syst* 2022;258:110036.
- Song X, Li J, Lei Q, Zhao W, Chen Y, Mian A. Bi-CLKT: Bi-graph contrastive learning based knowledge tracing. *Knowl-Based Syst* 2022;241:108274.
- Lu J, Lei L, Yang F, Shang L, Zeng X. Topology optimization of operational amplifier in continuous space via graph embedding. In: Bolchini C, Verbauwhede I, Vatajelu I, editors. *2022 design, automation & test in europe conference & exhibition*. IEEE; 2022, p. 142–7.
- Li J, Jin Y, Gao H, Qiang W, Zheng C, Sun F. Hierarchical topology isomorphism expertise embedded graph contrastive learning. 2023, CoRR abs/2312.14222.
- Rama-Maneiro E, Vidal JC, Lama M. Embedding graph convolutional networks in recurrent neural networks for predictive monitoring. *IEEE Trans Knowl Data Eng* 2024;36(1):137–51.
- Xia J. Knowledge-based course map generation for student performance prediction. In: Wang X, Sapino ML, Han W-S, Abbadi AE, Dobbie G, Feng Z, Shao Y, Yin H, editors. *Proceedings of the 28th international conference on database systems for advanced applications*. Cham: Springer Nature Switzerland; 2023, p. 748–52. http://dx.doi.org/10.1007/978-3-031-30678-5_67.
- Xia J, Tang Y, Zhao T, Li J. AuCM: Course map data analytics for Australian IT programs in higher education. In: Chen W, Yao L, Cai T, Pan S, Shen T, Li X, editors. *Proceedings of the 18th international conference on advanced data mining and applications*. Cham: Springer Nature Switzerland; 2022, p. 158–72. http://dx.doi.org/10.1007/978-3-031-22064-7_13.
- Xia J, Li M, Tang Y, Yang S. Course map learning with graph convolutional network based on AuCM. *World Wide Web* 2023;26(5):3483–502. <http://dx.doi.org/10.1007/s11280-023-01194-8>.
- Liu Y, Han T, Ma S, Zhang J, Yang Y, Tian J, He H, Li A, He M, Liu Z, Wu Z, Zhu D, Li X, Qiang N, Shen D, Liu T, Ge B. Summary of ChatGPT/GPT-4 research and perspective towards the future of large language models. 2023, <http://dx.doi.org/10.48550/ARXIV.2304.01852>, CoRR abs/2304.01852. URL <https://doi.org/10.48550/arXiv.2304.01852>.
- Kalyan KS. A survey of GPT-3 family large language models including ChatGPT and GPT-4. 2023, <http://dx.doi.org/10.48550/ARXIV.2310.12321>, CoRR abs/2310.12321.
- Yin H, Aryani A, Lambert G, Wu Z, Nambiar N, White M, Salvador-Carulla L, Sadiq S, Sojli E, Boddy J, Murray G, Tham WW. Leveraging artificial intelligence technology for mapping publications to sustainable development goals. *Array* 2025;27:100419. <http://dx.doi.org/10.1016/j.array.2025.100419>, URL <https://www.sciencedirect.com/science/article/pii/S2590005625000463>.

- [36] Cho K, van Merriënboer B, Gülçehre Ç, Bougares F, Schwenk H, Bengio Y. Learning phrase representations using RNN encoder-decoder for statistical machine translation. 2014, CoRR [abs/1406.1078](https://arxiv.org/abs/1406.1078).
- [37] Rumelhart DE, Hinton GE, Williams RJ. Learning internal representations by error propagation. *Nature* 1986;323(6088):533–6.
- [38] Hochreiter S, Schmidhuber J. Long short-term memory. *Neural Comput* 1997;9(8):1735–80.
- [39] Alagumalai S, Curtis DD. Classical test theory. Dordrecht: Springer Netherlands; 2005, p. 1–14.
- [40] Lord FM. Applications of item response theory to practical testing problems. Routledge; 2012.
- [41] Beck J, Stern M, Woolf BP. Using the student model to control problem difficulty. In: User modeling. Springer; 1997, p. 277–88.
- [42] Liu Y, Yang Y, Chen X, Shen J, Zhang H, Yu Y. Improving knowledge tracing via pre-training question embeddings. In: Bessiere C, editor. Proceedings of the twenty-ninth international joint conference on artificial intelligence. IJCAI.org; 2020, p. 1577–83.
- [43] Shen S, Huang Z, Liu Q, Su Y, Wang S, Chen E. Assessing student's dynamic knowledge state by exploring the question difficulty effect. In: Amigó E, Castells P, Gonzalo J, Carterette B, Culpepper JS, Kazai G, editors. SIGIR '22: the 45th international ACM SIGIR conference on research and development in information retrieval. ACM; 2022, p. 427–37. <http://dx.doi.org/10.1145/3477495.3531939>.
- [44] Dai H, Zhang Y, Yun Y, Shang X. An improved deep model for knowledge tracing and question-difficulty discovery. In: Pham DN, Theeramunkong T, Governori G, Liu F, editors. PRICAI 2021: trends in artificial intelligence - 18th Pacific rim international conference on artificial intelligence, PRICAI 2021, hanoi, Vietnam, November 8–12, 2021, proceedings, part II. Lecture notes in computer science, vol. 13032, Springer; 2021, p. 362–75.
- [45] Xiao B, Jiang H, Ma J, Zhang R. Tracing knowledge state with individualized ability and question difficulty. In: Hong W, Weng Y, editors. Computer science and education - 17th international conference, ICCSE 2022, ningbo, China, August 18–21, 2022, revised selected papers, part I. Communications in computer and information science, vol. 1811, Springer; 2022, p. 459–71.
- [46] Feng M, Heffernan NT, Koedinger KR. Addressing the assessment challenge in an intelligent tutoring system that tutors as it assesses. *J User Model User-Adapted Interact* 2009;19:243–66.
- [47] Abdelrahman G, Abdelfattah S, Wang Q, Lin Y. DBE-KT22: A knowledge tracing dataset based on online student evaluation. 2022, CoRR [abs/2208.12651](https://arxiv.org/abs/2208.12651).